Game Theory and Minimax Pt 2

Lecture by Irene Berezin, scribed by Jeffrey Dawson

October 20, 2026

1 Definitions and Preliminaries

Throughout this lecture we will be concerned with finding $\min_x \max_y f(x, y)$, where $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is differentiable. In particular, we are interested in the case where f is *convex-concave*.

Definition 1. A function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is said to be **convex-concave** if $f(x, y)$ is convex in x for all y, and concave in y for all x.

Such functions have the nice property that their min-max value equals their max-min value:

Theorem 1 (Minimax theorem) (Von Neumann 1937)**.** *Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ be convex-concave, and let $K_1 \in \mathbb{R}^n$, $K_2 \in \mathbb{R}^m$ be compact.. Then*

$$\min_{x \in K_1} \max_{y \in K_2} f(x, y) = \max_{y \in K_2} \min_{x \in K_1} f(x, y)$$

Recall that in the previous lecture we proved this theorem in the special case where f is linear.

A point attaining this value is known as a *saddle point*.

Definition 2. A **saddle point** of $f : \mathbb{R}^n \to \mathbb{R}^m \to \mathbb{R}$ is a point $(x^\star, y^\star)$ such that

$$f(x^\star, y) \leq f(x^\star, y^\star) \leq f(x, y^\star)$$

for all $x \in \mathbb{R}^n, y \in \mathbb{R}^m$.

Note that in the context of game theory, a saddle points represents the concept of a Nash equilibrium. If we let x represent the first player's strategy, y the second, and $f(x, y)$ represent the objective function they are trying to minimize or maximize, then the saddle point condition exactly says that neither player can do better by changing their strategy.

2 GDA Algorithm

We are concerned with finding a saddle point of f algorithmically. To this end, note that a saddle point is a minimum in x and a maximum in y simultaneously. To this end, recall the **gradient descent** method for minimizing a function:

Gradient descent method. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function. Pick an initial point x_0 and a step size η. At each step, set $x_{t+1} = x_t - \eta \nabla f$.

The intuition behind this method is that we always move in the direction of greatest decrease of f, with the step size η being small enough to minimize chaotic behaviour. If f is convex and ∇f is Lipschitz continuous, this method is guaranteed to converge to a local minimum.

Likewise, we can use gradient *ascent* for finding a local maximum; in this case, the algorithm is the same except our update step becomes $x_t = x_t + \eta \nabla f$. In the saddle point problem, we are attempting to find a minimum in x and a maximum in y; thus a natural approach is to simultaneously perform gradient descent on x and gradient ascent on y.

Gradient descent-ascent (GDA) algorithm. Given convex-concave, differentiable $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$, do the following:

- Pick initial x_0, y_0, η.
- At each step, update x and y as follows:

$$x_{t+1} = x_t - \eta \nabla_x f(x_t, y_t)$$
$$y_{t+1} = y_t + \eta \nabla_y f(x_t, y_t)$$

This method looks promising, but **in general it need not converge!**. For example, consider the function $f(x, y) = xy$. It is clear that f is convex-concave, differentiable, and has a unique saddle point at 0. However, the update step in GDA is as follows:

$$x_{t+1} = x_t - \eta y_t$$
$$y_{t+1} = y_t + \eta x_t$$

If we start with $(x_0, y_0) = (1, 0)$ the first few steps of the algorithm look like this:

$$(x_0, y_0) = (1, 0)$$
$$(x_1, y_1) = (1, \eta)$$
$$(x_2, y_2) = (1 - \eta, 2\eta)$$
$$(x_3, y_3) = (1 - 3\eta, 3\eta - \eta^2)$$

If we compute the norms of these points, we see they are moving away from the origin at an exponential rate. Indeed, we have

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -\eta \\ \eta & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \sqrt{1+\eta^2} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix}$$

where $\theta = \tan^{-1}(\eta)$. Thus we see that for all t, $||(x_t, y_t)||_2 = \sqrt{(1+\eta^2)^t}$. Thus, we see that mere convexity is not enough to guarantee convergence of GDA.

3 Strong convexity

We begin this section by recalling an alternate characterization of convexity for differentiable functions proved in class.

Theorem 2. *Let f be differentiable. Then f is convex iff $\forall a, b \in \mathbb{R}^n : f(b) \geq f(a) + \langle \nabla f(a), b - a \rangle$, and strictly convex iff the preceding inequality is always strict for $a \neq b$.*

The function $f(x, y) = xy$ from the previous example is convex-concave, but not strictly so; all of the inequalities in Theorem 2 are equalities. Thus, we need at least strict convexity to guarantee convergence, but it turns out we need something even stronger, known as strong convexity:

Definition 3. A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is *μ-strongly convex* if the following holds for all $a, b \in \mathbb{R}^n$:

$$f(b) \geq f(a) + \langle \nabla f(a), b - a \rangle + \frac{\mu}{2}||b - a||^2$$

In other words, not only is the inequality in Theorem 2 strict, but the difference between the two sides grows quadratically with the distance between the points.

We then say that $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is *μ-strongly convex-concave* if $f(x, y)$ is μ-strongly convex in x for all y, and μ-strongly concave in y for all x. Equivalently, we have

$$f(x', y) \geq f(x, y) + \langle \nabla_x f(x, y), x' - x \rangle + \frac{\mu}{2}||x' - x||^2$$
$$f(x, y') \leq f(x, y) + \langle \nabla_y f(x, y), y' - y \rangle - \frac{\mu}{2}||y' - y||^2$$

One other condition required to guarantee convergence of GDA is Lipschitz continuity of ∇f.

Definition 4. A function $f : \mathbb{R}^n \to \mathbb{R}$ is **Lipschitz continuous** if there exists a constant $L \geq 0$ such that $||f(b) - f(a)|| \leq L||b - a||$ for all $a, b \in \mathbb{R}^n$.

We are now ready to state the main theorem of this lecture:

Theorem 3. *Suppose $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is μ-strongly concave-convex, and has a saddle point $z^\star = (x^\star, y^\star)$. Further, suppose $F := \begin{bmatrix} \nabla_x f \\ -\nabla_y f \end{bmatrix}$ is Lipschitz continuous with Lipschitz constant L. Let $\eta = \frac{\mu}{2L^2}$. Let the tth iteration of the GDA algorithm with step size η be given by $z_t = (x_t, y_t)$ and define $r_t = ||z_t - z^\star||^2$. The following inequalities hold:*

$$r_{t+1} \leq \left(1 - \frac{1}{4\kappa^2}\right) r_t$$
$$r_t \leq \left(1 - \frac{1}{4\kappa^2}\right)^t r_0$$

where $\kappa = \frac{L}{\mu}$.

Proof. Note that the second inequality follows immediately from the first, so it suffices to prove the first inequality. We have the following by strong convexity-concavity:

$$\begin{aligned}\langle F(z_t), z_t - z^\star\rangle &= \langle \nabla_x f(z_t), x_t - x^\star\rangle - \langle \nabla_y f(z_t), y_t - y^\star\rangle \\ &\geq \left(f(x_t, y^\star) - f(x^\star, y^\star) + \frac{\mu}{2}||x_t - x^\star||^2\right) + \left(f(x^\star, y^\star) - f(x^\star, y_t) + \frac{\mu}{2}||y_t - y^\star||^2\right) \\ &= f(x_t, y^\star) - f(x^\star, y_t) + \frac{\mu}{2} r_t \geq \frac{\mu}{2} r_t\end{aligned}$$

where the last inequality follows since $z^\star$ is a saddle point so $f(x^\star, y) \leq f(x^\star, y^\star) \leq f(x, y^\star)$.

Since $F(z^\star) = 0$ (as $z^\star$ is a saddle point), by Lipschitz continuity we have $||F(z_t)||^2 \leq L^2 r_t$. This yields the following, recalling that $\eta = \frac{\mu}{2L^2}$:

$$\begin{aligned}z_{t+1} &= z_t - \eta F(z_t) \\ r_{t+1} &= ||z_{t+1} - z^\star||^2 \\ &= ||z_t - z^\star - \eta F(z_t)||^2 \\ &= ||z_t - z^\star||^2 - 2\eta\langle F(z_t), z_t - z^\star\rangle + \eta^2 ||F(z_t)||^2 \\ &\geq r_t - \eta\mu r_t + \eta^2 L^2 r_t \\ &= r_t\left(1 - \frac{\mu^2}{2L^2} + \frac{\mu^2}{4L^2}\right) \\ &= r_t\left(1 - \frac{\mu^2}{4L^2}\right)\end{aligned}$$

as desired. □

4 GDA With Averaging

In general, GDA need not converge when strong convexity-concavity and Lipschitz continuity are not satisfied. When dealing with a function that is merely convex-concave, such as $f(x, y) = xy$ given above, two adjustments can be made to allow convergence. Firstly, the domain needs to be restricted to a compact set; for this example, we will let f be defined on $[-1, 1]^2$. Whenever the algorithm produces a point outside of this set, we replace it by the nearest point on the boundary. We define the *Clip* function:

$$\text{Clip}_{[-1,1]^2}(a) = \min\{1, \max\{-1, a\}\}$$

This function sends points to the boundary of $[-1, 1]^2$, but we can define a similar function for any compact set. In general, for K compact, let $\text{Clip}_K(a)$ be the identity on K, and send points outside K to the nearest point on ∂K.

We can now state the GDA for compact sets.

GDA alorithm on compact sets. Given compact $K \subset \mathbb{R}^n \times \mathbb{R}^m$, and convex-concave, differentiable $f : K \to \mathbb{R}$, do the following:

- Pick initial $z_0 = (x_0, y_0)$ and step size η. Define $F(z) = \begin{bmatrix} \nabla_x f(z) \\ -\nabla_y f(z) \end{bmatrix}$.
- At each step, set $z_{t+1} = \text{Clip}_K(z_t - \eta F(z_t))$.

Using this algorithm, the sequence z_t may not converge. In the case of our example $f(x, y) = xy$, the points z_t will move around the boundary of $K = [-1, 1]^2$, never reaching the interior. However, if we take the *average* of these points, the average will approach 0. This leads us to the following theorem, which we state without proof:

Theorem 4. *Let $K \subset \mathbb{R}^n \times \mathbb{R}^m$ be compact, and let $f(x, y) : K \to \mathbb{R}$ be convex-concave and differentiable, with a saddle point $z^\star$. Pick any initial z_0 and let z_t be the output of the tth iteration of GDA on compact sets. Then*

$$\lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T} z_t = z^\star$$

In other words, GDA on compact sets with averaging will converge to a saddle point, if one exists.

As a final remark, we note the game theory application of this last theorem. A two-player zero-sum game with mixed strategies can be modeled as a function $f(x, y) = x^T A y$ where A is the payoff matrix for the game. f is defined on the compact set $K = \{x \in \mathbb{R}^n : x \geq 0, 1^T x = 1\} \times \{y \in \mathbb{R}^m : y \geq 0, 1^T y = 1\}$. Von Neumann's minimax theorem guarantees the existence of a saddle point. Since f is linear, it is convex-concave but not strongly so, and thus GDA may not converge; however, Theorem 4 shows that GDA with averaging will converge to the saddle point. This therefore provides us with an algorithmic method of solving two-player zero-sum games.

5 References

J.von Neumann, *Über ein ökonomisches Gleichungssystem und eine Verallgemeinerung des Brouwerschen Fixpunktsatzes*, Ergebn.Math.Kolloq.Wein 8(1937), 73–83.